

Programme	BSc. Computer Science				
Course Title	Predictive Modelling				
Type of Course	Vocational Minor				
Semester	VIII				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Basic Mathematical Concepts 2. Basic Statistics				
Course Summary	Predictive Modeling gives undergraduate students a solid foundation in predictive analytics techniques essential for data-driven decision-making. The course covers key topics such as correlation, covariance, linear regression, multiple regression, polynomial regression, logistic regression, and time series analysis and forecasting.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Apply correlation and covariance analysis to assess relationships between variables.	Ap	P	Problems/ Projects
CO2	Implement linear, multiple, and polynomial regression models to predict outcomes from numerical data.	Ap	P	Problems/ Projects
CO3	Employ time series analysis techniques to identify trends, and seasonal patterns, and make accurate forecasts.	An	P	Problems/ Projects
CO4	Evaluate model performance and interpret results to inform business decisions.	An	P	Analysis of reports and case studies
CO5	Utilize logistic regression to classify categorical outcomes and make data-driven decisions.	Ap	P	Projects
CO6	Acquire proficiency in building predictive models using real-world datasets	U	C	Assignments/ Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48)	Marks (70)
I	Correlation & Covariance		8	12
	1	Data types or levels of measurement- Nominal, ordinal, interval and ratio	2	
	2	Covariance sample and population, sign and magnitude of covariance,	1	
	3	The covariance matrix, Covariance vs Correlation	2	
	4	Measures of Correlation, Simple correlation	1	
	5	Partial correlation and Multiple correlations	2	
II	Regression Techniques		12	16
	6	Simple linear regression	2	
	7	Basics of fitting and residual analysis	2	
	8	Multiple linear regression	2	
	9	Gauss Markov theorem	2	
	10	Least Squares Method, ordinary least squares, weighted least squares	2	
	11	Polynomial regression	2	
III	Logistics Regression		11	14
	12	Basics of Logistic regression	2	
	13	Logistic regression with binary predictor	2	
	14	Odds ratio, z-statistic, p-values	3	
	15	Confidence intervals	2	
	16	Logistic regression with categorical predictors	2	
IV	Time Series analysis and forecasting		16	28
	17	Components of time-series, additive and multiplicative models	3	
	18	Methods for measurement of trends	2	
	19	Methods for measurement of seasonal fluctuations	3	
	20	Forecasting, Autocorrelation	2	
	21	ARIMA Model	3	
	22	ARMA Model	3	
V	Open Ended Module: Assignments, Case study		12	
	1. Provide real-world examples to understand the relationships between variables in data analysis using covariance, correlation 2. Using real examples, understand the difference between different types of correlation. 3. Provide examples for nominal, ordinal, interval, and ratio data types 4. Provide examples of how linear regression is used in various fields such as economics, finance, healthcare, and engineering to analyze relationships between variables and make predictions.		4	
	5. Analyze the relationship between different variables using multiple linear regression. Eg: Health Care analytics: patients' demographics, lifestyle factors, and medical history using multiple linear regression. 6. Analyze the relationship between different variables using logistic regression. Eg: Predict the risk of developing a certain disease (e.g.,		8	

	diabetes, cancer) using logistic regression with binary predictors such as genetic markers, lifestyle factors, and medical history. 7. Analyze the time series model using ARIMA/ ARMA model		
--	---	--	--

Mapping of COs with PSOs and POs:

	PS O1	PS O2	PS O3	PSO 4	PS O5	PS O6	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6
CO 1	3	-	3	-	-	-						
CO 2	3	-	3	2	-	2						
CO 3	3	-	3	2	-	2						
CO 4	-	-	2	-	-	2						
CO 5	3	-	3	2	-	2						
CO 6	-	-	2	-	-	2						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓	✓	✓
CO 2	✓	✓	✓	✓
CO 3	✓	✓	✓	✓
CO 4	✓		✓	✓

CO 5	✓			✓
CO 6	✓			✓

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Final Exam

References:

1. Fan, Jianqing, et al. Statistical Foundations of Data Science. United States, CRC Press, 2020.
2. Hilbe, Joseph M. Practical Guide to Logistic Regression. United States, CRC Press, 2016.
3. Nielsen, Aileen. Practical Time Series Analysis: Prediction with Statistics and Machine Learning. United States, O'Reilly Media, 2019.
4. Fundamentals of Mathematical Statistics. United Kingdom, Sultan Chand & Sons, 2020.